

Matej Olexa

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Master's student in AI and Cybersecurity with 2.5 years of research experience. Combines machine learning, embedded systems, and security expertise. Delivers both research prototypes and production software.

EDUCATION

Brno University of Technology, Faculty of Information Technology Brno, Czech Republic
Master's in Information Technology and Artificial Intelligence (Cybersecurity specialization) 2025 - Present

Brno University of Technology, Faculty of Information Technology Brno, Czech Republic
Bachelor's in Information Technology 2022 - 2025

– Bachelor's Thesis: "Monitoring of Bluetooth Low Energy" (Grade: A)

RESEARCH EXPERIENCE

LLM Security Researcher (RedHat/IBM Joint Research Project) September 2025 - Present
Security@FIT, Brno University of Technology Brno, Czech Republic

- Primary researcher on detection of knowledge editing methods (ROME, MEMIT) in LLMs as part of the [RedHat/IBM LLM Forensics](#) research project under PhD supervision. Paper submitted to **ACM CCS**. Research focuses on model integrity, adversarial robustness, and identifying knowledge manipulation patterns.
- Built [Latium](#) – an open-source framework for causal tracing, weight intervention, and per-layer spectral analysis of edited LLM weights, designed for SOTA models.

BLE Security Researcher September 2023 - August 2025
Security@FIT, Brno University of Technology Brno, Czech Republic

- **Data Acquisition:** Co-engineered distributed monitoring system using ESP32 microcontrollers (C/ESP-IDF) for large-scale BLE traffic analysis. Implemented synchronized data collection pipeline (Python) with millisecond-precision timestamping across multiple hardware sensors.
- **Machine Learning Detection:** Designed and implemented neural network architectures (MLPs, 1D CNNs in PyTorch) to detect BLE connection events in real-time from passive advertising data. Achieved 96% F1 score on resource-constrained embedded hardware.
- **Recognition:** Awarded Professional Recognition at [EXCEL@FIT 2025](#). Presented at [SantaCrypt 2025](#). Published dataset: github.com/hujon/ble-ard. Thesis: [BUT Link](#) / [GitHub](#).

WORK EXPERIENCE

Security & Software Developer April 2025 - December 2025
Guardians.cz Brno, Czech Republic

- Architected NIS2 compliance automation engine from scratch to production deployment. Built full-stack solution (Python backend, SvelteKit/TypeScript frontend) with Bitbucket CI/CD pipeline. System now in active client use, automating compliance assessment beyond previous industry capabilities.
- Consulted on DevSecOps practices including container security, SSDLC implementation, and CI/CD security integration.

TECHNICAL SKILLS

Programming Languages: C++ (C++17/20, STL), C, Python, TypeScript, SQL

Machine Learning & Data: PyTorch, NumPy, Pandas

DevOps & Tools: Docker, CI/CD (Bitbucket Pipelines), Git, Linux (Fedora), Bash

Embedded Systems: ESP32 (ESP-IDF, MicroPython), Arduino

Security Tools: gdb (gef), pwntools, Binary Ninja

Web Frameworks: SvelteKit, Next.js

Languages: Slovak (Native), Czech (Near-Native), English (Fluent), German (Basic)

COMPETITIONS & PROJECTS

CyberApocalypse CTF 2025

HackTheBox

- Team MVP, 166th/8130 teams globally. Authored technical writeups for binary exploitation challenges: [Contractor](#), [Strategist](#).

CyberHeroes Academy SOC Exercise

Binary Confidence, 2025

- Led SOC analyst role in 7-person team. First team in program history to successfully defend against red team attacks, achieving record-breaking performance.

Second Foundation Hackathon - 2nd Place

2025

- Engineered high-performance energy trading backend in Go for real-time trading simulation (4-person team). [GitHub](#)

BSEC 2026 Hackathon - 1st Place

Brno University of Technology, 2026

- Built a multi-agent influencer content platform in a 12-hour sprint (4-person team). Features a swarm of 4 role-specialized AI agents (trend scout, producer, analyst, QA reviewer) that debate and vote on content direction, an anti-hallucination research pipeline with web search and location verification, and automated generation of platform-ready outputs (YouTube script + thumbnail, Instagram post, TikTok reel + video). [GitHub](#)

More projects

olexamatej.github.io/index-projects.html